Self-Study 1

- The below questions are from Problem Solving section.
- Write individual answers in your books.
- Use the tabular format to identify the Input, Process and Output.
- Use line numbers to write down the process clearly.

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1. Preparing a letter which can be posted
2. Preparing a cup of tea
3. Dividing 40 pages and 80 pages books from a parcel of books between two siblings (say Judi and Sam)
4. Divide 100 toffees which are taken from a bag among 20 people equally.
5. Do the question 4 again using an alternative algorithm/process
6. Class teacher marking the attendance of 20 students using the register
    a. All 20 students may not be in the class
7. Calculate the number of words in a paragraph.
8. Calculate the addition of two numbers when user input two numbers.
    a. Addition = number 1 + number 2
9. Calculate the volume of a box when 3 dimensions are given (length, width and height)
    a. Volume = length x width x height
10. Calculate the surface area of a Cube when the length of an edge is given by user.
    a. Area of a cube = 6 x length^2
11. Calculate the area of a Circle when the Radius is given by user. You can treat the π value as a constant value as 3.1415
    a. Area of circle = π x radius^2
12. Calculate the area of a Cone when Radius and Height are given by user.
    a. Area of a cone = π x radius (radius +$\sqrt{height^2 + radius^2}$)
13. Ask for two numbers and find the largest number out of them.
14. Figure out whether a given number is negative or positive.
15. Figure out whether a given number is even or odd.